

CPSC 304 Project Cover Page

Milestone #: 3

Date: Mar 1, 2026

Group Number: 45

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above.

In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

Project Summary:

Our project models a structured healthy meal planning system that enables users to manage recipes, track nutritional information, and filter meals based on dietary needs and allergies. The system connects users with dietitian designed recipes and supports ingredient level nutrient tracking, substitution options, and health based classifications.

Week 1 (March 7 – March 13): Database Setup and Initial Query Implementation

Focus: Implement full CRUD functionality along with Selection and Projection queries on the Oracle server side. No graphical interface is included at this stage.

Each team member is responsible for developing two queries, and all queries will be consolidated into a single shared SQL file.

Every query must execute successfully in SQL*Plus using meaningful sample data

Member	Responsibility	Deliverables
Esther	Write Query 1 (Insert) & Query 4 (Selection)	Successfully running SQL script with commented examples
Dali	Write Query 2 (Update) & Query 5 (Projection)	Successfully running SQL script with commented examples
Jiyang	Write Query 3 (Delete) & Finalize init.sql (5 inserts)	Complete init.sql with test data for all tables

Checkpoint (Mar 13): All basic queries return correct output in SQL*Plus

Code pushed to GitHub.

Week 2 (March 14 – March 20): Backend API Development, Intermediate Queries and debug all SQL

Focus: Build Node.js endpoints and connect to Database.

Advanced query logic and collaborative testing.

Each team member develops one aggregation query, while the division query is implemented collectively.

Member	Responsibility	Deliverables
Esther	Implement Query 6 (Join) and develop the corresponding backend API routes	New API endpoints added in server.js (Endpoints)
Dali	Implement Query 7 (Aggregation) and build the basic frontend HTML page	index.html including navigation structure and basic data table layout
Jiyang	Setup backend environment (.env mgmt) & Integration testing	Stable server connection; frontend fetching data

Week 3 (March 21 – March 27):: Advanced Logic & Dynamic Interaction

Focus: Tackle complex queries and frontend rendering.

Member	Responsibility	Deliverables
Esther	Implement Query 9 (Nested Aggregation) & Query 10 (Division)	Stress-tested complex SQL logic and API endpoints
Dali	Write Query 8 (Having) & Implement dynamic data rendering	Webpage dynamically displaying SQL query results
Jiyang	Implement frontend filter forms (e.g., Protein sliders/inputs)	User-friendly input interface for queries

Checkpoint (Mar 27):

All 10 queries can be triggered and displayed via the frontend interface.

Week 4 (March 28 – April 2):: Final Debugging & Submission

Focus: System-wide testing and code cleanup.

Member	Responsibility	Deliverables
All	"Scorched Earth" testing; ensure all 10 queries work	Bug-free final project code
Esther	Optimize SQL performance; ensure Foreign Key Cascades work	Optimized database schema
Dali	Final CSS polishing; ensure responsive web design	High-quality User Interface (UI)
Jiyang	Prepare M4 deliverables; clean up redundant GitHub files	Final project URL submitted to Canvas

Checkpoint (Apr 2):

M4 submitted; repository is in "Production Ready" state.

Summary of division:

Member	Workload %	Core Contribution
Esther	33.30%	Complex SQL (Division/Nested Aggregation) & Backend API architecture.
Dali	33.30%	System-wide UI/CSS design & Dynamic data rendering/interaction.
Jiyang	33.30%	Environment setup, basic CRUD logic & Input form integration.

Potential Challenges:

A key challenge is ensuring smooth integration between frontend, backend, and database components developed by different team members. To reduce inconsistencies, we will adopt standardized naming conventions and response formats.

Another challenge involves implementing complex SQL logic, including aggregation and division queries. We will address this by testing queries independently and reviewing them collaboratively before integration.

Acknowledgment About Use Of AI

We did not use AI for Milestone3.